Swaminathan Sundar

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Summary

- Ph.D. candidate in Chemical Engineering at Texas A&M University specializing in multiscale modeling, mathematical optimization, and data-driven decision-making for energy and manufacturing systems.
- 5+ years of experience with Python-based modeling and analytics (Pyomo, Pandas, NumPy, PyTorch), nonlinear and mixed-integer programming (NLP/MILP), stochastic optimization, and model predictive control (MPC/NMPC).
- First-author publications in *Computers & Chemical Engineering* applying optimization, TEA/LCA, and digital twins to large-scale planning and operations problems (e.g., DOE funded CCUS supply chains).

Technical Skills

Programming: Python (Pyomo, Gurobi, Pandas, NumPy, PyTorch, Keras, TensorFlow, scikit-learn), MATLAB, C++
Operations Research & Optimization: Stochastic & robust optimization (two-stage stochastic programming), nonlinear and mixed-integer programming (NLP/MILP), multi-parametric programming, decision-making under uncertainty, supply chain & scenario/risk analysis, heuristic optimization, graph theory, dynamic and multiscale high-fidelity modeling, model predictive control (MPC/NMPC), NN-based forecasting (w/ uncertainty quantification), surrogate modeling, time-series clustering, global sensitivity analysis (Sobol), model-based design of experiments (MBDoE).
Modeling, Simulation & Solvers: Pyomo, Gurobipy, GAMS, MATLAB, LINGO, AMPL, Pyomo.DAE, gPROMS, Aspen Plus, COMSOL, Microsoft Excel; Gurobi, IPOPT, CPLEX, PPOPT, BARON, ANTIGONE, GLPK.
Sustainability & Analytics: Techno-economic analysis (TEA), static & dynamic LCA, supply-chain modeling, data wrangling and analysis in Python (Pandas/NumPy), and visualization in Python.

Experience

Texas A&M Energy Institute, College Station, TX
Graduate Research Assistant **Aug 2021 - Present**

- Designed a generalized multiscale digital-twin framework linking unit operations, plant-level operations, and supply-chain decisions for emerging process technologies, demonstrated on algae-based CO_2 utilization and designed to be transferable to energy, chemical, and manufacturing networks.
- Built high-fidelity photobioreactor digital twins in Python/Pyomo with transport, reaction, and biological kinetics, coupled with global parameter estimation, global sensitivity analysis, and model-based design of experiments (MBDoE) to obtain robust, well-identified models for decision support.
- Developed forecast-aware nonlinear model predictive control (NMPC) formulations embedding exogenous forecasts (e.g., solar irradiance) into closed-loop optimization, combining dynamic models and time-series data for real-time operational decision making.
- Created a generalized TEA/LCA framework using a state–task network representation that automatically generates mixed-integer optimization models with cost and emissions KPIs for technology selection, capacity expansion, inventory management, and logistics planning.
- Extended this framework to a two-stage stochastic optimization model that separates long-term investment decisions from scenario-dependent operations, enabling risk-aware planning under uncertain prices and demand profiles.
- Developed a dynamic LCA methodology with time-dependent climate characterization factors for short-lived climate pollutants and integrated it with optimization-based planning in agricultural and energy-transition case studies.
- Led modeling efforts for the DOE-funded Continuous Algae-based Carbon Capture and Utilization (CACCU) project, using algae-based systems as detailed case studies of the generalized digital-twin and TEA/LCA frameworks.

Carnegie Mellon University, Pittsburgh, PA
Graduate Research Assistant **Aug 2019 - Dec 2020**

- Modeled and optimized a photo-catalytic CO_2 utilization process in Pyomo, integrating Langmuir–Hinshelwood kinetics, annular fluid dynamics, and radiation models.
- Achieved 25× higher CO_2 conversion by optimizing reactor length and recycle configuration; performed sensitivity analysis to identify key levers for design and operation.

Selected Projects

NLP for Review Classification with Transformers, Texas A&M University, Aug 2023 – Dec 2023

- Built an NLP pipeline on 174,000 Yelp reviews using Python (NLTK, PyTorch).
- Implemented custom positional encoding and a Transformer-based classifier; performed hyperparameter tuning and achieved 83% validation and 82.9% test accuracy.

Metaheuristic Optimization for Capacitated Vehicle Routing (CVRP), Texas A&M University, Jan 2022 – May 2022

- Designed a metaheuristic (2-opt and λ-interchange with Iterated Local Search) for CVRP to generate near-optimal routes.
- Benchmarked performance across problem sizes and analyzed scalability and runtime behavior.

Optimization and Quantum Annealing for Particle Track Reconstruction, Carnegie Mellon University, Aug 2020 – Oct 2020

- Formulated particle-track reconstruction from LHC data as both a mixed-integer optimization problem (Pyomo) and a QUBO model.
- Implemented simulated and quantum annealing-based solvers; evaluated solution quality and scaling with number of particles.

Education

Texas A&M University — College Station, TX
Doctor of Philosophy in Chemical Engineering; GPA: 3.7/4.0 — *Aug 2021 – Aug 2026 (expected)*
Advisor: Prof. Efstratios N. Pistikopoulos
Dissertation: *Multiscale Modeling and Optimization Framework for Digital Twin Development Applications*

Relevant Coursework: *Nonlinear Programming, Advanced Process Optimization, Data Mining and Analysis, Machine Learning, Deep Learning*

Carnegie Mellon University — Pittsburgh, PA
Master of Science in Chemical Engineering; GPA: 3.7/4.0 — *Aug 2019 – Dec 2020*
Advisors: Profs. Debangsu Bhattacharyya and Chrysanthos Gounaris

Relevant Coursework: *Advanced Process Systems Engineering, Product and Supply Chain Optimization, Quantum Integer Programming, Data Science in Chemical Engineering*

Teaching, Leadership, and Honors

- Teaching Assistant: Process Dynamics & Control; Fluid Operations in Chemical Engineering; Advanced Process Optimization.
- Ranked among the top 20 teams globally in the 2024 Shell.ai Hackathon Challenge.
- Peer reviewer, *Computers & Chemical Engineering*, *Bioresource Technology* and *Biotechnology Advances*.
- Contributed to multiple grant proposals, including the successful $26 million NSF CURB Engineering Research Center initiative.
- Vice President, Chemical Engineering Master's Student Association (ChEMSA), CMU: organized the first ChEMSA Research Symposium; supported Master's students during COVID-19.
- Business Development Executive, AIESEC Navi Mumbai: co-organized Youth Speak Forum 2018; helped raise 200k for the chapter.

Publications

1. Swaminathan Sundar, Rahul Kakodkar, and Efstratios N Pistikopoulos. Techno-economic analysis and life cycle assessment of a novel algae-based ccus technology. *Computers & Chemical Engineering*, page 109409, 2025

2. Swaminathan Sundar, Vincent Xu, Bin Long, Susie Dai, Joshua Yuan, Yinjie J. Tang, and Efstratios N. Pistikopoulos. From dynamic modeling to optimal experiments: An mbdoe framework for scale-up of algae based ccus process. In preparation, 2026

3. Swaminathan Sundar, Vincent Xu, Dustin Kenefake, Yinjie J. Tang, and Efstratios N. Pistikopoulos. Forecast-aware nonlinear model predictive control of photobioreactors under solar irradiance uncertainty. In preparation, 2026